Important update: Arm Announces End of Life Timeline for Mbed. This site will be archived in July 2026. **Read the full announcement.** (https://os.mbed.com/blog/entry/Important-Update-on-Mbed/)

NUCLEO-F746ZG

STM32 Nucleo-144 development board with STM32F746ZGT6 MCU, supports Arduino, ST Zio and morpho connectivity

To compile a program for this board using Mbed CLI, use **nucleo_f746zg** as the target name.

Board Partner

life.augmented

(/teams/ST/)

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A world leader in providing the semiconductor solutions that make a positive contribution to people's lives, both today and in the future.

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Overview

The STM32 Nucleo-144 board provides an affordable and flexible way for users to try out new concepts and build prototypes with the STM32 microcontroller, choosing from the various combinations of performance, power consumption and features. The ST Zio connector, which is an extension of Arduino™ Uno, provides access to more peripherals and ST morpho headers make it easy to expand the functionality of the Nucleo open development platform with a wide choice of specialized shields. The STM32 Nucleo-144 board does not require any separate probe, as it integrates the ST-LINK/V2-1 debugger/programmer and it comes with the STM32 comprehensive software HAL library, together with various packaged software examples, as well as a direct access to the ARM®mbed™online resources.

Microcontroller features

- STM32F746ZGT6 in LQFP144 package
- ARM®32-bit Cortex®-M7 + FPU + Chrom-ART™ Accelerator
- 216 MHz max CPU frequency
- VDD from 1.7 V to 3.6 V
- 1 MB Flash
- 320 KB SRAM
- GPIOs (168) with external interrupt capability
- 12-bit ADCs with 24 channels (3)
- 12-bit DAC channels (2)
- USART/UART (8)
- I2C (4)
- SPI (6)
- Advanced-control Timer (2)
- Low-power Timer (1)
- General Purpose Timers (12)
- Watchdog Timers (2)
- CAN 2.0B active (2)
- SAI (2)
- Random Generator (TRNG for HW entropy)
- USB 2.0 OTG HS
- USB 2.0 OTG FS
- Ethernet

Nucleo features

- Two types of extension resources
  - Arduino Uno Revision 3 connectivity
  - STMicroelectronics Morpho extension pin headers for full access to all STM32 I/Os
- On-board ST-LINK/V2-1 debugger/programmer with SWD connector
  - Selection-mode switch to use the kit as a standalone ST-LINK/V2-1
- Flexible board power supply
  - USB VBUS or external source (3.3 V, 5 V, 7 - 12 V)
  - Power management access point
- Three User LEDs
- Two push buttons: USER and RESET
- USB re-enumeration capability: three different interfaces supported on USB
  - Virtual Com port
  - Mass storage (USB Disk drive) for drag'n'drop programming
  - Debug port

Buy Now (https://www.st.com/en/evaluation-tools/nucleo-f746zg.html#sample scroll)

Mbed Enabled

- Advanced
- Baseline

Mbed OS support

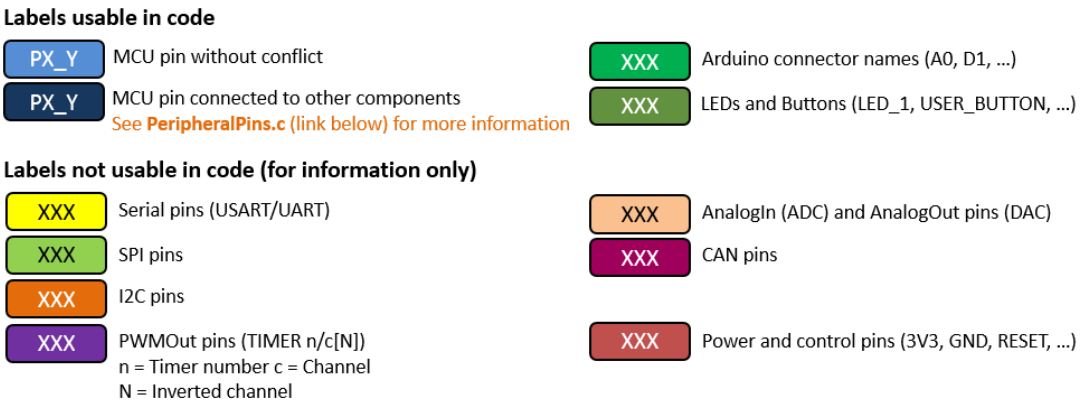
- Mbed OS 2
- Mbed OS 5.10
- Mbed OS 5.11
- Mbed OS 5.12
- Mbed OS 5.13
- Mbed OS 5.14
- Mbed OS 5.15
- Mbed OS 6.0
- Mbed OS 6.1
- Mbed OS 6.10
- Mbed OS 6.11
- Mbed OS 6.12
- Mbed OS 6.13
- Mbed OS 6.14
- Mbed OS 6.15
- Mbed OS 6.2
- Mbed OS 6.3
- Mbed OS 6.4
- Mbed OS 6.5
- Mbed OS 6.6
- Mbed OS 6.7
- Mbed OS 6.8
- Mbed OS 6.9

Example programs

- Debug port
- Ethernet 10/100Mbps
- USART

Board pinout

Pins Legend



You can find more details on the available pins and labels in the **PeripheralPins.c** and **PinNames.h** files.

These files can be found in:

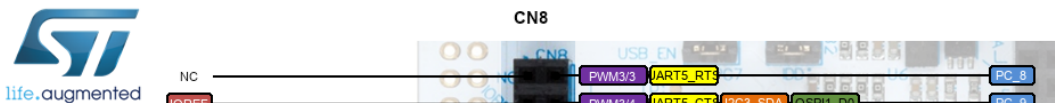
- ARMmbed/mbed-os repository on GitHub (**up-to-date** version, used with **mbed CLI commands**)

https://github.com/ARMmbed/mbed-os/blob/master/targets/TARGET_STM/TARGET_STM32F7/TARGET_STM32F746xG/TARGET_NUCLEO_F746ZG/ (https://github.com/ARMmbed/mbed-os/blob/master/targets/TARGET_STM/TARGET_STM32F7/TARGET_STM32F746xG/TARGET_NUCLEO_F746ZG/)

- mbed-dev library in developer.mbed.org (source files of the mbed library used on **mbed compiler IDE**)

https://developer.mbed.org/users/mbed_official/code/mbed- dev/file/default/targets/TARGET_STM/TARGET_STM32F7/TARGET_STM32F746xG/TARGET_NUCLEO_F746ZG/ (https://developer.mbed.org/users/mbed_official/code/mbed- dev/file/default/targets/TARGET_STM/TARGET_STM32F7/TARGET_STM32F746xG/TARGET_NUCLEO_F746ZG/)

Zio and Arduino-compatible headers



programs

Mbed 2 deprecated

Sensors_Reader (/teams/ST/code/Sensors_Read...

70 (/teams/ST/code/Sensors_Reader/}shortlog)

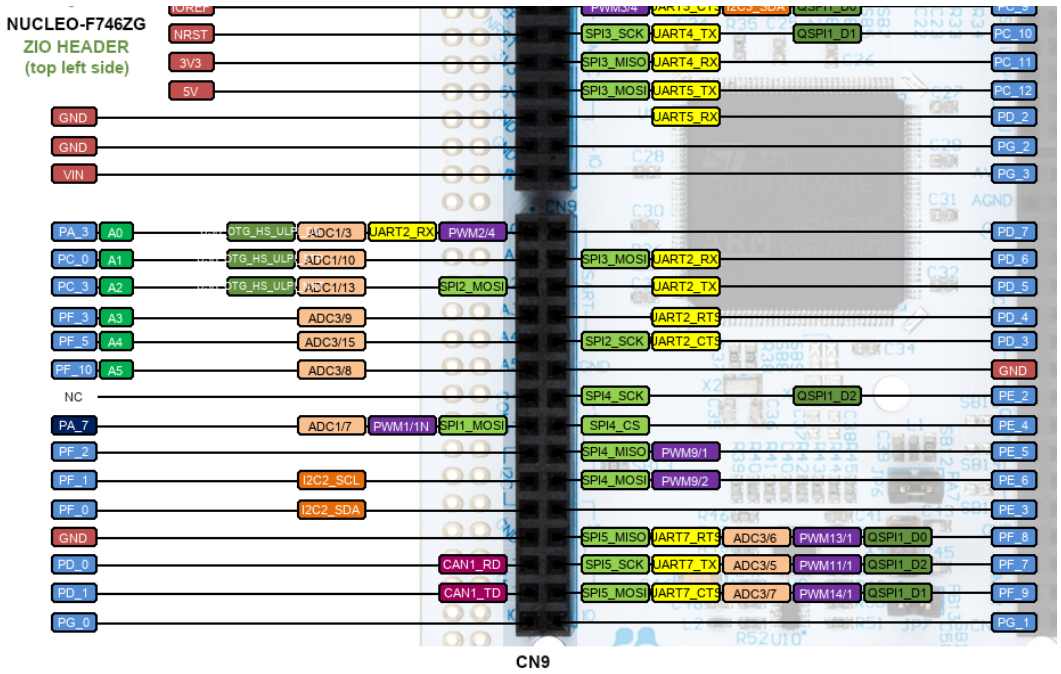
2335 (/teams/ST/code/Sensors_Reader/)

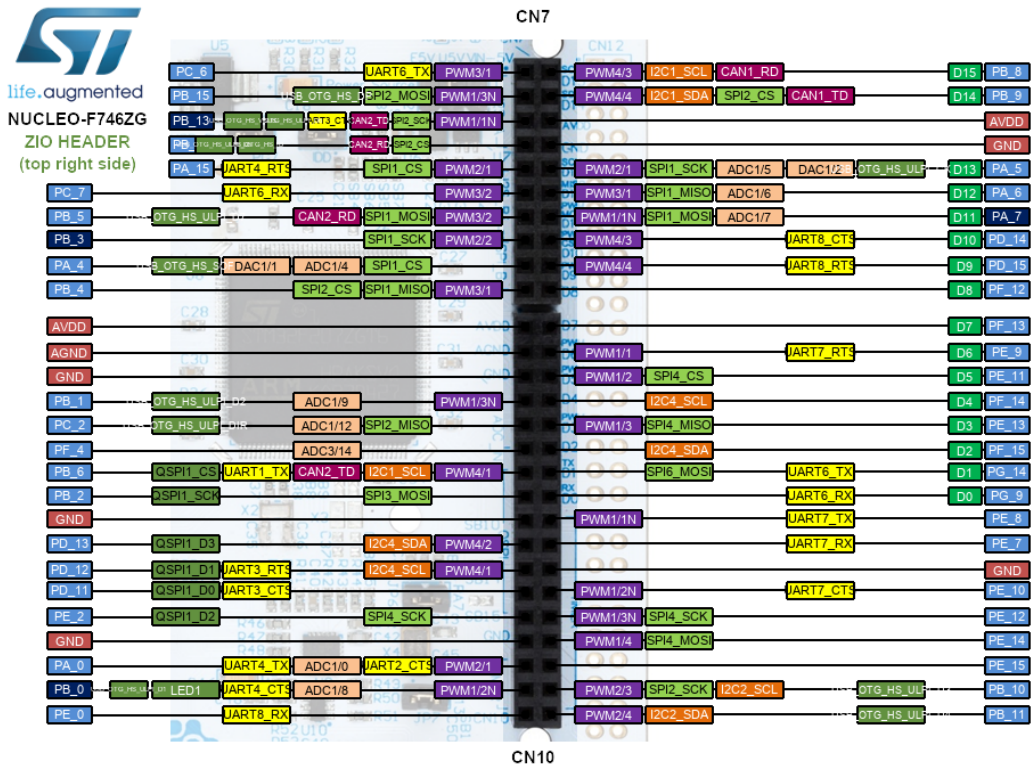
"Sensors Reader" Sample Application for X-NUCLEO-IKS01A1 Expansion Board

arduino (/search/?q=arduino), ExpansionBoard (/search/?q=ExpansionBoard), HTS221 (/search/?q=HTS221), LIS3MDL (/search/?q=LIS3MDL), lps25h (/search/?q=lps25h), LSM6DS0 (/search/?q=LSM6DS0), LSM6DS3 (/search/?q=LSM6DS3), Nucleo (/search/?q=Nucleo), X-NUCLEO-IKS01A1 (/search/?q=X-NUCLEO-IKS01A1)

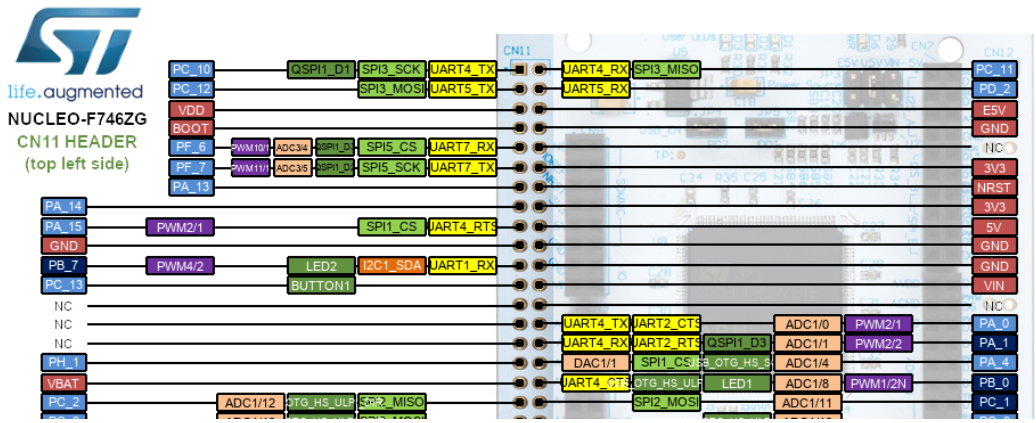
Last updated: 24 Mar 2017 (24 Mar 2017)

Mbed OS mbed-os-example-mbed5-blinky (/





CN11 CN12 headers



teams/mbed-os-examples/code/mbed-os-example-mbed5-blinky/)

✔ Featured

107 (/teams/mbed-os-examples/code/mbed-os-example-mbed5-blinky/}shortlog)

195500 (/teams/mbed-os-examples/code/mbed-os-example-mbed5-blinky/)

This is a very simple guide, reviewing the steps required to get Blinky working on an Mbed OS platform.

Last updated: 22 Nov 2019 (22 Nov 2019)

Mbed 2 deprecated

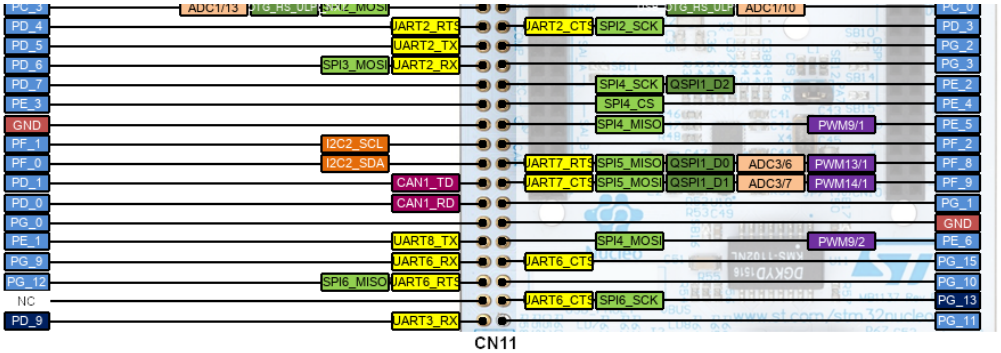
STM32_ADC_... (/teams/ST/code/STM32_ADC_...

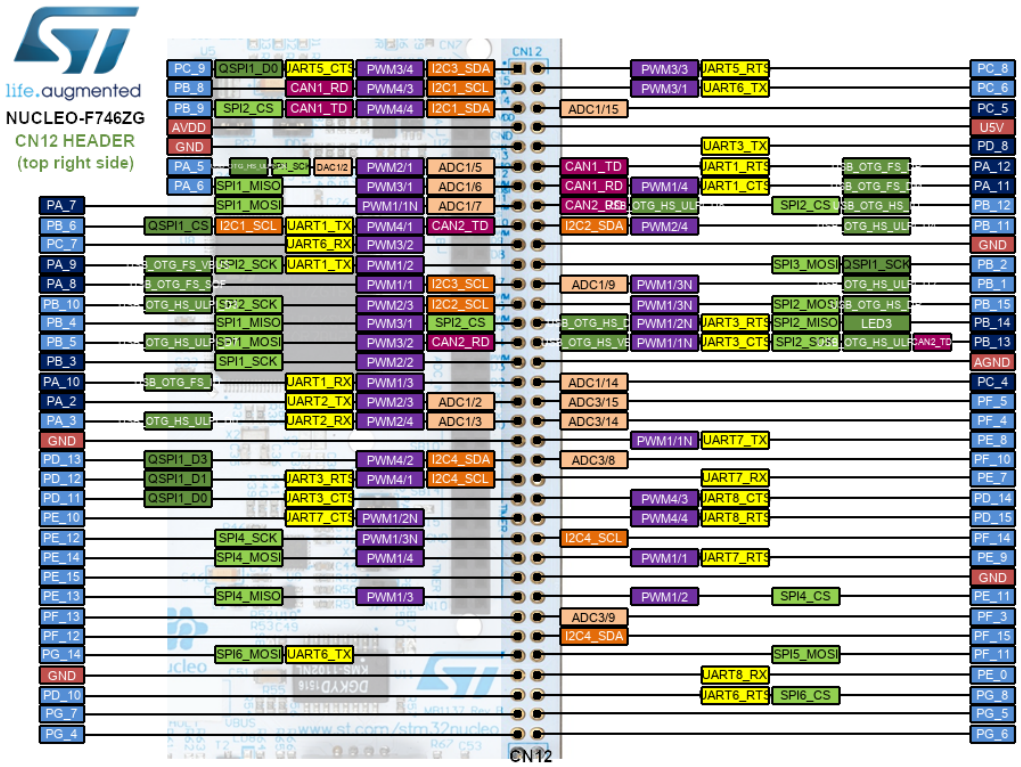
9 (/teams/ST/code/STM32_ADC_InternalCh shortlog)

49840 (/teams/ST/coc STM32_ADC_InternalCh

ADC internal channels read example.

Internal Channels (/search/?q=Internal Channels), stm32 (/search/?q=stm32),





Getting started

This video shows how to get started with ARM mbed Integrated Development Environment using STM32 Nucleo platform:

temperature sensor (/search/?q=temperature sensor), VBAT (/search/?q=VBAT)

Last updated: 17 Aug 2017 (17 Aug 2017)

Mbed OS example-Ethernet-mbed-Cloud-connect (/teams/ST/code/example-Ethernet-mbed-Cloud-connect/)

15 (/teams/ST/code/example-Ethernet-mbed-Cloud-connect/} shortlog)
878 (/teams/ST/code/example-Ethernet-mbed-Cloud-connect/)

Superseded by https://os.mbed.com/teams/ST/code/pelion-example-common/

cloud (/search/?q=cloud), ethernet (/search/?q=ethernet), stm32 (/search/?q=stm32)

Last updated: 15 Oct 2018 (15 Oct 2018)

Mbed OS pelion-example-common (/teams/ST/code/pelion-example-

Supported shields

ST X-NUCLEO boards

See Matrix of tested boards (https://developer.mbed.org/teams/ST/wiki/Matrix-of-tested-boards).

Other Non-ST boards

See here (https://developer.mbed.org/teams/ST/wiki/Supported-shields).

Technical references

For more information, please refer to:

- STM32F746ZG microcontroller (http://www.st.com/web/catalog/mmc/FM141/SC1169/SS1858/LN1902/PF260873?s_searchtype=partnumber)
- NUCLEO-F746ZG board (http://www.st.com/web/catalog/tools/FM116/CL1620/SC959/SS1532/LN1847/PF262631?s_searchtype=partnumber)
- SDK changes log (/teams/ST/wiki/SDK-changes-log)

Known limitations

The following section describes known limitations of the platform. Note that general issues are tracked into the mbed repository (https://github.com/mbedmicro/mbed) available on GitHub.

- Conflict between Ethernet and SPI pins on Arduino D11

See HERE (https://os.mbed.com/teams/ST/wiki/Nucleo-144pins-ethernet-spi-conflict)

common/)

26 (/teams/ST/code/pelion-example-common/}shortlog)

4698 (/teams/ST/code/pelion-example-common/)

Mbed OS Device Management example for various ST boards.

device management (/search/?q=device management), Pelion (/search/?q=Pelion), ST (/search/?q=ST), ST Discovery (/search/?q=ST Discovery), ST Nucleo (/search/?q=ST Nucleo)

Last updated: 27 Mar 2019 (27 Mar 2019)

CMSIS support

Find documentation, software examples and the CMSIS Board Support Pack.

NUCLEO-F746ZG on keil.arm.com (https://www.keil.arm.com/boards/stmicroelectronics-nucleo-f746zg-revb-fbabf0d/)

Tips and Tricks

Find more information in ST WIKI pages (https://os.mbed.com/teams/ST/wiki/Special:Allpages).

Buy Now (https://www.st.com/en/evaluation-tools/nucleo-f746zg.html#samplebuy-scroll)

You need to log in (/account/login/?next=/platforms/ST-Nucleo-F746ZG/) to post a discussion

Questions

2 answers
(/questions/85884/Temperature-rating-for-Nucleo-F746zg-0C-/) ✔
Temperature rating for Nucleo-F746zg (0C to 85C?) (/questions/85884/Temperature-rating-for-Nucleo-F746zg-0C-/) (/users/dan_le/) Dan Le (/users/dan_le/)
6 years, 10 months ago (Tue 14 May 2019 16:31)

NUCLEO-F746ZG (/questions/tag/NUCLEO-F746ZG)

0 answers
(/questions/79028/Using-PB7-as-a-serial-pin-instead-of-LED/) Using PB7 as a serial pin instead of LED2 (/questions/79028/Using-PB7-as-a-serial-pin-instead-of-LED/) (/users/mag2017/) Chamika Mihiran (/users/mag2017/)
8 years, 6 months ago (Thu 14 Sep 2017 10:47)

NUCLEO-F746ZG (/questions/tag/NUCLEO-F746ZG)

2 answers
(/questions/78241/How-change-pin-function/) ✔ [closed] How change pin function ? (/questions/78241/How-change-pin-function/) (/users/CecileRapoutet/) Cécile Rapoutet (/users/CecileRapoutet/)
8 years, 9 months ago (Wed 14 Jun 2017 14:20)

NUCLEO-F746ZG (/questions/tag/NUCLEO-F746ZG)

0 answers
(/questions/78240/How-change-pin-function/) How change pin function ? (/questions/78240/How-change-pin-function/) (/users/CecileRapoutet/) Cécile Rapoutet (/users/CecileRapoutet/)
8 years, 9 months ago (Wed 14 Jun 2017 14:20)

6 years, 7 months ago (Wed 14 Jun 2017 14:26)

NUCLEO-F746ZG (/questions/tag/NUCLEO-F746ZG)

0 answers
(/questions/77537/I2C/) I2C (/questions/77537/I2C/) (/users/jos4/)
jos rennies (/users/jos4/)
9 years ago (Fri 24 Mar 2017 10:15)

stm32-F746ZG (/questions/tag/stm32-F746ZG)

See more related questions (/questions/related/72/147/ST-Nucleo-F746ZG/)

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(http://forums.mbed.com/) (http://blog.mbed.com/)

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